

Viscosity Measurement: Falling Ball, Saybolt, and Rotary Viscometers

Student Handout

Fluid Power Systems and Internal Combustion Engines

AGRICULTURAL MECHANIZATION AND BUSINESS · CLEMSON UNIVERSITY

Name	_____	Date	_____
Group	_____	Bench	_____

1 Objectives

- To become familiar with viscosity measurement methods.
- To measure viscosity of hydraulic fluids with falling ball viscometer, Saybolt viscometer and rotary viscometer.
- To convert one set of viscosity units to another set of units.
- To analyze the effect of temperature on viscosity of hydraulic fluid.

2 Core Principle

Viscosity is a measure of a fluid's resistance to flow. None of the three instruments on this bench measures that resistance directly. Each one measures something else, how fast a sphere sinks, how long a fixed volume takes to drain, how hard a rotor must be driven, and then infers viscosity from a model. The model is where the physics lives, and the model can fail while the instrument still gives you a perfectly repeatable number.

Two quantities carry the name viscosity and they are not interchangeable. *Dynamic* (or absolute) viscosity μ measures resistance to shear and is reported in Pa·s or centipoise. *Kinematic* viscosity ν is dynamic viscosity divided by density and is reported in m²/s or centistokes. Density is the only bridge between them. An ISO hydraulic grade such as ISO 46 refers to kinematic viscosity in centistokes at 40 °C; a Stokes' law result is dynamic viscosity in Pa·s. Quoting one where the other is expected is the most common error in this laboratory.

WHAT YOU SHOULD BE ABLE TO SAY AT THE END

Which of your three measurements you trust most, and why. That answer depends on the Reynolds number you compute in Section 5.6, not on which instrument looked most expensive.

3 Safety

- Fluids heated to 50 to 60 °C will burn skin. Use gloves and tongs on the Saybolt cup and the rotary beaker.
- Wipe every spill at once. Oil on a hard floor is a fall hazard.
- Do not heat any fluid above the temperature stated in the procedure.
- Recover the steel sphere with the magnet. Never invert the falling ball tube.
- Used fluid goes in the marked waste container, never down the sink.
- Report a cracked tube or a leaking cup to the instructor before proceeding.

4 Constants and Conversions

Record these once and use them throughout.

$$g = 9.81 \text{ m/s}^2 \quad \cdot \quad 1 \text{ in} = 2.54 \text{ cm} \quad \cdot \quad 1 \text{ m} = 100 \text{ cm} \quad \cdot \quad 1000 \text{ g} = 1 \text{ kg} \quad \cdot \quad 1 \text{ m}^3 = 1000 \text{ L} = 10^6 \text{ cm}^3$$

$$1 \text{ poise} = 0.1 \text{ Pa}\cdot\text{s} = 100 \text{ cP} \quad \cdot \quad 1 \text{ cP} = 0.001 \text{ Pa}\cdot\text{s} \quad \cdot \quad 1 \text{ Pa}\cdot\text{s} = 1000 \text{ cP} = 1000 \text{ mPa}\cdot\text{s} \quad \cdot \quad \text{cSt} = \text{cP} / \rho [\text{g/cm}^3]$$

5 Part I: Falling Ball Viscometer

One of the oldest ways of measuring the viscosity of a fluid is to measure how fast a sphere falls through it. A sphere released in a viscous fluid accelerates for a short distance and then settles at a constant *terminal velocity*, where the drag force exactly balances the sphere's weight less its buoyancy. If the sphere falls quickly the fluid has low viscosity and little resistance to flow; if it falls slowly the fluid has high viscosity. Stokes' law relates that settling velocity to viscosity, rearranged below to solve for viscosity.

$$\mu = \frac{2 \Delta \rho g r^2}{9 v} \quad (1)$$

- μ absolute (dynamic) viscosity, Pa·s
- $\Delta \rho$ density of sphere minus density of fluid, kg/m³
- g acceleration due to gravity, 9.81 m/s²
- r radius of sphere, m
- v velocity of sphere, m/s

RELEASE THE SPHERE ABOVE THE START MARK

Equation (1) applies only to a sphere already at terminal velocity. If you release the sphere at the START mark it is still accelerating as it crosses it, your timed velocity is too low, and your viscosity comes out too high. Release it at the top of the tube and let it settle before it reaches the mark. The bench sphere needs about 0.25 m of travel to reach terminal velocity in a motor oil and about 0.5 m in an ATF, so the START mark must be at least that far below the release point.

THE WHOLE STATION IN ONE MULTIPLICATION

Once the sphere, the marks and the fluid density are fixed, Equation (1) collapses to $\mu = K t_{\text{avg}}$ with $K = 2 \Delta \rho g r^2 / (9L)$. For the bench sphere and marks, K is about 0.68 Pa·s per second for both bench fluids. Compute K once, then every fall time becomes a viscosity in one step. Section 5.6 reduces the same way: $Re = \rho_f L(2r) / (K t_{\text{avg}}^2)$, about $14/t_{\text{avg}}^2$ on this bench.

5.1 Steel sphere

Measure the diameter of the sphere with a micrometer and its mass on the balance. Compute the volume from Equation (2) and the density from Equation (3).

$$V = \frac{4}{3} \pi r^3 \quad (2)$$

$$\rho = \frac{m}{V} \quad (3)$$

Mass of sphere: _____ g = _____ kg

Diameter D : _____ in = _____ cm = _____ m Radius r : _____ m

Volume V : _____ m³ Sphere density ρ_s : _____ kg/m³

5.2 Fluid density

Fill a beaker with 50 mL of each fluid and weigh it. Record the temperature with a thermometer. Divide the measured mass in grams by the volume in millilitres to obtain the density, then carry it through to kg/m³.

Quantity	Fluid 1	Fluid 2
Fluid name		
Temperature, °C		
Beaker volume, mL		
Fluid mass, g		
ρ , g/mL		
ρ , g/L		
ρ , kg/m ³		

Table 1. Fluid identification and density.

5.3 Fall time and velocity

Measure the distance between the two marked points on the viscometer. Hold the sphere with the magnet, release it from the top of the tube, and time its travel between the two marks. Repeat four times and average. Four trials are taken because a single stopwatch reading carries a reaction-time error of roughly a tenth of a second, which on a two-second fall is a five percent error on its own.

Distance between marks L : _____ m

Fluid	t_1 , s	t_2 , s	t_3 , s	t_4 , s	t_{avg} , s	v , m/s
1.						
2.						

Table 2. Fall times over the measured distance.

$$t_{\text{avg}} = \frac{t_1 + t_2 + t_3 + t_4}{4} \quad v = \frac{L}{t_{\text{avg}}} \quad (4)$$

5.4 Viscosity and unit conversion

Compute $\Delta\rho$, then apply Equation (1). Convert the result to centipoise and to centistokes. Show the substituted numbers, not only the answer.

Fluid 1: $\Delta\rho = \underline{\hspace{2cm}} - \underline{\hspace{2cm}} = \underline{\hspace{2cm}} \text{ kg/m}^3$
 $\mu = \underline{\hspace{2cm}} \text{ Pa}\cdot\text{s} = \underline{\hspace{2cm}} \text{ cP} = \underline{\hspace{2cm}} \text{ cSt}$

Fluid 2: $\Delta\rho = \underline{\hspace{2cm}} - \underline{\hspace{2cm}} = \underline{\hspace{2cm}} \text{ kg/m}^3$
 $\mu = \underline{\hspace{2cm}} \text{ Pa}\cdot\text{s} = \underline{\hspace{2cm}} \text{ cP} = \underline{\hspace{2cm}} \text{ cSt}$

5.5 Uncertainty in the fall time

Report the spread of your four readings, not only their average. The range divided by the average, expressed as a percentage, is a defensible first estimate of the uncertainty in your velocity, and therefore in your viscosity, since μ is inversely proportional to v .

Fluid 1: range = _____ s range/ t_{avg} = _____ % $\rightarrow$ $\mu = \underline{\hspace{2cm}} \pm \underline{\hspace{2cm}} \text{ Pa}\cdot\text{s}$
 Fluid 2: range = _____ s range/ t_{avg} = _____ % $\rightarrow$ $\mu = \underline{\hspace{2cm}} \pm \underline{\hspace{2cm}} \text{ Pa}\cdot\text{s}$

5.6 Is Stokes' law valid for your measurement?

Equation (1) is derived for *creeping flow*, meaning the viscous forces on the sphere overwhelm the inertial ones. The Reynolds number measures that ratio, and Stokes' law requires it to be much less than one.

$$Re = \frac{\rho_f v (2r)}{\mu} \quad (5)$$

Fluid 1: $Re =$ _____ Circle one: valid ($Re < 1$) · marginal (1 to 10) · invalid ($Re \geq 10$)

Fluid 2: $Re =$ _____ Circle one: valid ($Re < 1$) · marginal (1 to 10) · invalid ($Re \geq 10$)

The Reynolds number above uses the viscosity you are testing, so when Stokes' law fails the inflated μ makes Re look smaller than it is. Recompute it with the rotary reading for the same fluid at the same temperature (Part III). That is the honest check, and it is the value to use in Questions 5 and 10.

Fluid 1: Re with rotary $\mu =$ _____ Fluid 2: Re with rotary $\mu =$ _____

DO NOT DISCARD A NUMBER BECAUSE THE MODEL FAILED

If your Reynolds number is above one, report the viscosity you calculated *and* state that Stokes' law does not strictly apply to it. A result reported with its limitation is worth more than a result quietly deleted. Question 5 asks you to say which direction the error runs.

6 Part II: Saybolt Viscometer

The Saybolt viscometer measures fluid flow through a capillary of very small diameter. Before filling the vessel, secure the rubber stopper at the base. The container must be filled just to overflowing for accurate and consistent results, because the height of fluid above the orifice sets the driving pressure. Position the beaker underneath and begin timing at the instant the stopper is removed. Record the time in Saybolt Universal Seconds required for 60 mL to collect in the beaker.

Heat 60 mL of each fluid to 50 °C, pour it into the viscometer, and record the collection time.

Quantity	Fluid 1	Fluid 2
Fluid name		
Temperature, °C		
Collection time, SUS		
Equation used, (6) or (7)		
ν , cSt		
ρ , g/cm ³		
μ , cP		

Table 3. Saybolt measurements and conversions.

Convert the collection time to kinematic viscosity with the expression appropriate to your reading, then to dynamic viscosity using the density you measured in Table 1.

$$\text{For } 32 < \text{SUS} < 100 : \quad \nu [\text{cSt}] = 0.226 \text{ SUS} - \frac{195}{\text{SUS}} \quad (6)$$

$$\text{For } \text{SUS} > 100 : \quad \nu [\text{cSt}] = 0.220 \text{ SUS} - \frac{135}{\text{SUS}} \quad (7)$$

$$\mu [\text{cP}] = \nu [\text{cSt}] \times \rho [\text{g/cm}^3] \quad (8)$$

THE TWO EXPRESSIONS MEET

Both Equation (6) and Equation (7) return 20.65 cSt at SUS = 100, so the pair is continuous and the choice of branch at the boundary does not matter. Below 32 s neither applies; if your reading is that short, the fluid is too thin for this instrument and you should say so rather than force a number out of Equation (6).

7 Part III: Rotary Viscometer

A rotary viscometer comes closest to reporting the absolute viscosity of a fluid directly. A rotor immersed in the sample is driven at a set speed, and the torque required to maintain that speed is proportional to the fluid's viscosity. Sensors inside the instrument measure that torque and the viscosity is displayed on the screen. The measurement is easier to perform and generally more accurate than the other two methods, which is what makes it a useful reference against your falling-ball result.

Identify the components of the instrument and become familiar with the values shown on the display. Fill a beaker with 500 mL of fluid and heat it to 60 °C. Record the temperature and the displayed viscosity in mPa·s at 5 °C intervals as the sample cools, and convert each reading to Pa·s using Equation (9).

$$\mu [\text{Pa} \cdot \text{s}] = \frac{\mu [\text{mPa} \cdot \text{s}]}{1000} \quad (1 \text{ Pa} \cdot \text{s} = 1000 \text{ mPa} \cdot \text{s}) \quad (9)$$

Record the rotor and the speed before you take any reading. The torque the instrument senses depends on the rotor diameter, its immersed height, the clearance to the beaker wall and the rotation speed, so a viscosity value taken on this bench is only reproducible by someone else if all of them are written down. A reading quoted without its spindle and speed is not a measurement, it is a number.

Rotor diameter D : _____ mm Immersed height H : _____ mm Speed: _____ rev/min
 Beaker inner diameter: _____ mm Fluid depth: _____ mm Annular gap $(B - D)/2$:
 _____ mm

Trial	Set temp. °C	Measured temp. °C	Viscosity mPa·s	Viscosity Pa·s
1	20			
2	25			
3	30			
4	35			
5	40			
6	45			
7	50			
8	55			
9	60			

Table 4. Rotary viscometer readings. Fluid type: _____

7.1 What the instrument computes

The display gives you a viscosity directly, but it is worth knowing what stands behind that number, because it explains why the spindle and the speed must be recorded. Treat the rotor as a cylinder of radius R_i turning at angular velocity ω inside a stationary beaker of inner radius R_o , immersed to a depth H . The instrument holds the speed constant and measures the torque M required to maintain it.

$$\omega = \frac{2\pi N}{60} \quad \dot{\gamma} = \frac{2\omega R_o^2}{R_o^2 - R_i^2} \quad (10)$$

$$\tau = \frac{M}{2\pi R_i^2 H} \quad \mu = \frac{\tau}{\dot{\gamma}} \quad (11)$$

Combining the two gives the viscosity in terms of the measured torque alone:

$$\mu = \frac{M (R_o^2 - R_i^2)}{4\pi \omega H R_i^2 R_o^2} \quad (12)$$

N	rotation speed, rev/min	ω	angular velocity, rad/s
R_i	rotor radius, m	R_o	beaker inner radius, m
H	immersed height, m	M	torque on the rotor, N·m
$\dot{\gamma}$	shear rate, s ⁻¹	τ	shear stress, Pa

Every quantity in Equation (12) except M and N is fixed once the spindle and beaker are chosen, so the whole expression collapses to $\mu = KM/N$, where K is a pure geometric constant. That constant is the spindle factor printed in the instrument's tables, and it is the reason a reading is meaningless unless the spindle and the speed are recorded alongside it.

ASSUMPTIONS BEHIND EQUATION (12)

The fluid is Newtonian, the flow is laminar, the rotor is fully immersed and turning coaxially in the beaker, and the torque contributed by the top and bottom faces of the rotor is neglected. Equation (12) is exact for a long cylinder in a close-fitting beaker and becomes an approximation as the gap widens.

Compute for one of your readings: $\omega =$ _____ rad/s $\dot{\gamma} =$ _____ s⁻¹ $M =$ _____ N·m

7.2 Temperature plot

Plot temperature on the horizontal axis and viscosity on the vertical axis. Sketch the curve here during the laboratory period, then produce the same plot in a spreadsheet and submit it with your report. Label both axes with units.



20 °C

Temperature

60 °C

8 Analysis Questions

- Which of your two fluids has the greater resistance to flow? Cite the specific measured quantity that establishes this, not the calculated viscosity alone.

2. Your falling-ball result is a dynamic viscosity and your Saybolt result begins as a kinematic viscosity. Explain why the Saybolt method yields kinematic viscosity directly, given that the fluid is driven through the orifice by its own weight.

3. Compare the viscosity you obtained for the same fluid by the falling-ball method and by the rotary viscometer at the same temperature. Express the disagreement as a percentage of the rotary value.

4. List two experimental causes of the disagreement in Question 3 that are independent of the validity of Stokes' law.

5. Using your Reynolds numbers from Section 5.6, state whether Stokes' law was applicable to each fluid. For any fluid where it was not, state the *direction* of the resulting error: at Reynolds numbers above one, drag on the sphere is larger than Equation (1) assumes, so does your computed viscosity read high or low? Justify from Equation (1).

6. Propose one change to the falling-ball apparatus that would bring the Reynolds number into the valid range, and estimate the factor by which it would reduce Re . Note that v is proportional to r^2 , so Re is proportional to r^3 .

7. Describe the shape of your temperature-viscosity curve. Is a straight line an adequate description of it across the whole range? What does your answer imply about quoting a single viscosity value for a hydraulic fluid without stating a temperature?

8. A hydraulic system designed around a fluid at 40 °C is started on a winter morning at 5 °C. Using the trend in your data, argue whether the pump inlet or the actuator is the more likely place for trouble, and say what physically goes wrong there.

9. An engine oil is labelled 15W-40 and a hydraulic fluid is labelled ISO 46. State which viscosity family each grading system refers to and at what temperature, and explain why the two labels cannot be compared directly.

10. Of your three measurements of the same fluid, which do you trust most? Defend the choice using your uncertainty estimate from Section 5.5 and your Reynolds numbers from Section 5.6, not the reputation of the instrument.

9 Interactive Simulator

An interactive version of all three instruments is available in the browser. It runs the same equations printed in this handout, animates the sphere falling at the velocity your own timings imply, drains the Saybolt cup over your own SUS reading, and plots your rotary data as you enter it. It computes the Reynolds number live and labels the Stokes regime, so you can see directly how sphere size moves a measurement in and out of validity. Use it to check your arithmetic and to explore fluids the bench does not stock. It does not replace the bench measurements.

<https://alibulentkoc.github.io/viscosity-measurement-lab/lab/viscosity-lab.html>

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